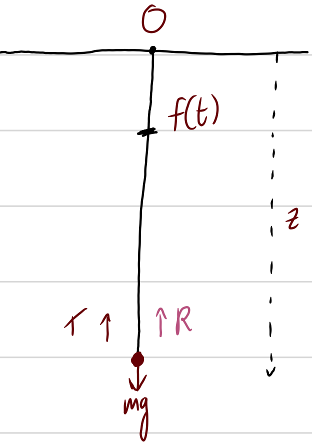
Forced and Damped Oscillations



elastic string has modulus λ and natural length l_0.
assume the string is never slack

There is an additional force R dampening the ocsillation.
This force acts proportional to v.

- Gravity (mg)
- Tension in string $\left[\frac{\lambda}{l_0}(z - f(t) - l_0)\right]$
- Damping force $(km\dot{z})$

Equation of motion: $F = m\ddot{z} = mg - km\dot{z} - \frac{\lambda}{l_0}(z - f(t) - l_0)$

$$m\ddot{x} = \frac{\lambda}{l_0}(x - f(t)) - km\dot{x}$$

$$\ddot{x} + 2k\dot{x} + \omega_0^2 x = F(t)$$

Auxillary equation: $m^2 + 2km + \omega_0^2 = 0 \Rightarrow m = \frac{-2k \pm \sqrt{4k^2 - 4\omega_0^2}}{2} = -k \pm \sqrt{k^2 - \omega^2}$

Then $x = e^{pt}\left[A\cos(qt) + B\sin(qt)\right] = e^{-kt}\left[A\cos(\sqrt{k^2-\omega^2}) + B\sin(\sqrt{k^2-\omega^2})\right] + P.I$